Ethernet Technology and Virtualization

Types of connections

- Wired
  - Copper
  - Fibre Optic
- Wireless

Copper cables

- The Network Access layer of the TCP/IP networking model converts the message into an appropriate signal for the network media before placing the message onto the actual network.
- These cables are inexpensive network cables that are very easy to implement within buildings, office spaces, and homes.

There are two types of copper cables that are used within networks:

- Unshielded Twisted Pair (UTP)
- Shielded Twisted Pair (STP)

Unshielded Twisted Pair (UTP)

- These cables are very common and used in almost any network.
- It does not contain a protective coating to prevent the actual conductors from absorbing **electromagnetic interference** (**EMI**).
- It can absorb EMI from nearby devices and other electrical wires and corrupts bits of transmitted message.
- It referred to as *twisted pair* as a total of 8 individual wires that have a unique color coating are twisted in a total of 4 pairs.
- It is cheap compared to shielded twisted pair (STP).
- It's not recommended to implement UTP cables in areas or zones that contain a lot of machinery that emits EMI.

The following figure shows a UTP cable:

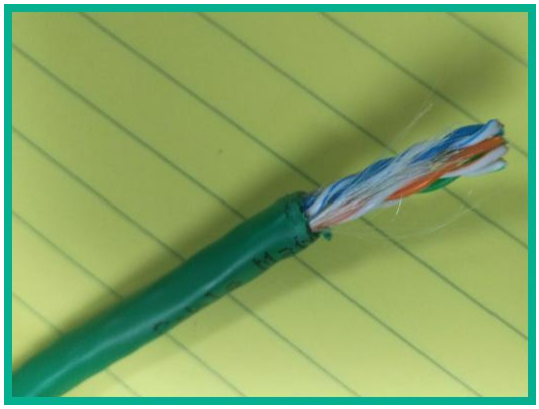


Fig. – UTP cable

Shielded Twisted Pair (STP)

- It is a type of twisted cable that contains protective shielding around the twisted conductors within the outer jacket.
- It is used to prevent EMI from entering the copper conductors within the cable.
- It is a bit more expensive.
- It is implemented within environments that have a lot of EMI.
- The following figure shows an STP cable:

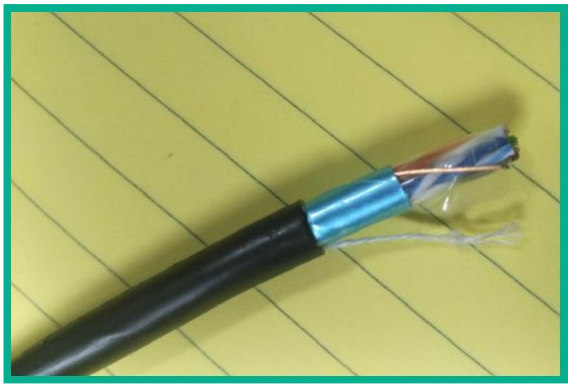


Fig. – STP cable

Plenum-rated cable

- Both the UTP and STP cables are made using Polyvinyl Chloride (PVC), a material that releases toxic/harmful fumes to humans when burnt. These are known as non-plenum-rated cables.
- It's highly recommended to implement plenum-rated cables within the plenum spaces within buildings when needed.
- These cables contain a plenum-grade outer jacket to prevent fire from burning the conductors.
- There is a pair of separator spines to keep each pair of twisted cables separate from other pairs.
- These are more expensive compared to non-plenum-rated cables.
- These are less toxic.

The following diagram shows an example of a plenum-rated cable:

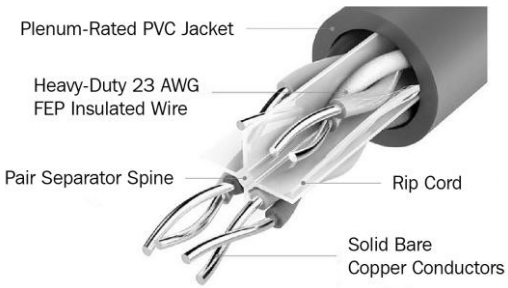


Fig. – Plenum-rated cable

Cabling standard

- The **Telecommunication Industry Association** (**TIA**) created a cabling standard that's used by networking professionals within the industry.
- These cables are described as a **category** (**Cat**) cable followed by a number.
- Each category defines how the cable can be applied to a network, its support speed, such as bandwidth, and the maximum length for data transmission.
- The following is a list of various types of category cables that are commonly found within organizations:
  - **Cat 3**: Supports speeds up to 10 Mbps with a maximum distance of 100 meters
  - **Cat 5**: Supports speeds up to 100 Mbps with a maximum distance of 100 meters
  - **Cat 5e**: Supports speeds up to 1 Gbps with a maximum distance of 100 meters
  - **Cat 6**: Supports speeds up to 1 Gbps with a maximum distance of 100 meters
  - **Cat 6a**: Supports speeds up to 1 Gbps and 10 Gbps with a maximum distance of 100 meters

- **Cat 7**: Supports speeds up to 1 Gbps and 10 Gbps with a maximum distance of 100 meters
- **Cat 8**: Supports speeds up to 40 Gbps with a maximum distance of 30 meters

- The ends of the twisted pair cables are terminated by various types of **Registered Jack** (**RJ**) connectors.
- For CAT 3 cable, an RJ-11 connector is used on both ends of the cable. It contains 4 pins; these are commonly used on traditional landline telephone systems.

The following figure shows a CAT 3 cable with an RJ 11 connector on its end:

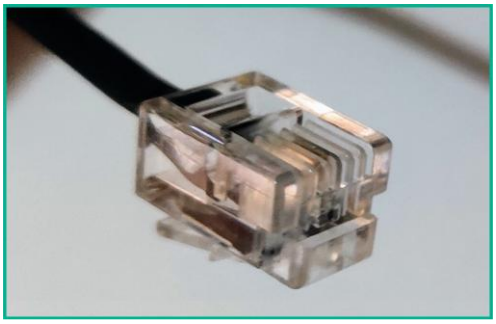


Fig. – Cat 3 with an RJ 11 connector

- The RJ 45 connector is an 8-pin connector that is used on most modern Ethernet cables, such as CAT 5 and above.

The following figure shows an Ethernet cable with an RJ 45 connector:

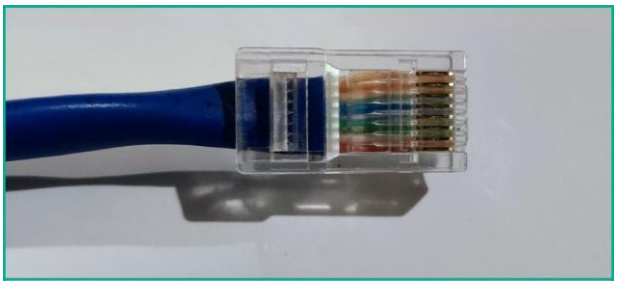


Fig. – Ethernet cable with an RJ 45 connector

Coaxial Cable

The following are the various layers of a coaxial cable, from the outer to inner layers:



Fig. – Coaxial cable

- **Protective coating**: Made using PVC and protects the internal media from external elements and environmental factors
- **Braided shielding**: Prevents EMI from reaching the copper conductor

- **Foil shielding**: Prevents **Radio Frequency Interference** (**RFI**) from reaching the copper conductor
- **Dielectric Insulator**: An insulator to separate the copper conductor and the braided shielding and to keep the copper conductor at the center of the coaxial cable
- **Copper conductor**: It is used to transmit the electrical signals along the cable.
- The following three types of connectors are used on coaxial cables:
  - **F-pin connector**
  - **Bayonet Neill-Concelman** (**BNC**) connector
  - **T type connector**, which is used to interconnect three coaxial cables together

Fiber optic cables

- These cables transmit data using pulses of light that are sent down a thin core of plastic or glass, which is surrounded by cladding.
- The combination of core and cladding allows for the light to be transmitted through the process of either total internal reflection or continuous refraction of the light.
- The following figure shows a fiber optic cable:

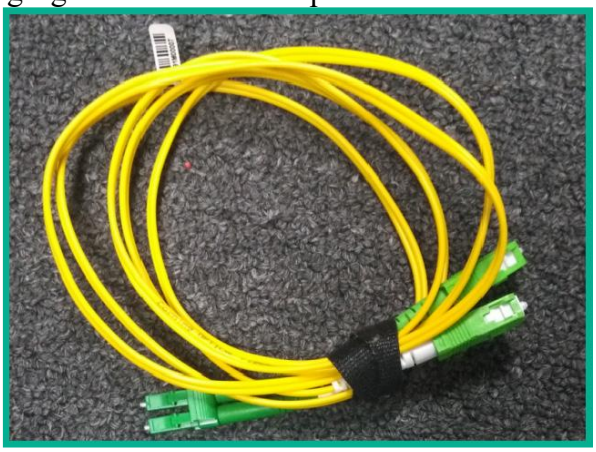


Fig. – Fiber optic cable

Advantages

- It allows faster throughput of data over the cable since it uses **Light Emitting Diodes** (**LED**) or lasers to transmit data in the form of light.
- Fiber optic cables can run for many kilometers before needing a repeater.
- Fiber optic cables are immune to EMI.

Disadvantages

- These are more expensive compared to copper cables.
- These are very fragile and easy to break as the core is either glass or plastic.

Structure of Fiber Optic Cable

The following diagram shows the structure of a fiber optic cable:

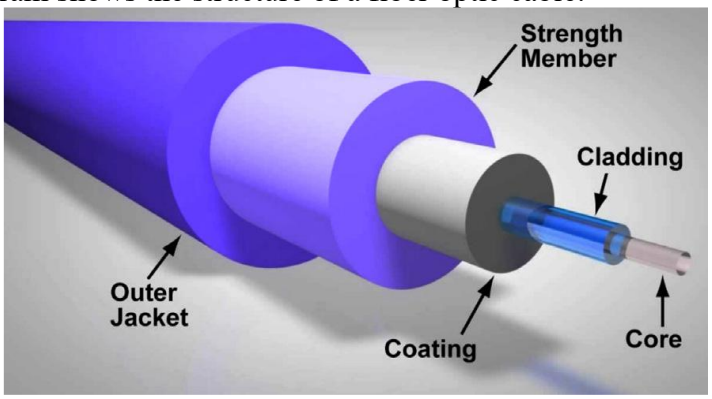


Fig. – Fiber optic cable structure

- It contains multiple layers of protection to prevent damage to the core.
- Each layer, such as the outer jacket, strength member, coating, and cladding, shields the core (glass or plastic) from external and environmental factors.

Categories of fiber cables:

1. single-mode
2. multimode.

Single-mode fiber

- It transmits light only in one mode through the fiber (in a direction parallel to the fiber).
- These cables consist of a core with a diameter that is quite small concerning the diameter of the cladding. For example, 9/125 fiber, which means the core is 9 **micrometers** or **microns** (**µm**) in diameter, while the cladding is 125 µm.
- Light through SMF can consist of multiple frequencies, but all of these frequencies follow a single path through the fiber.
- It is used for fibers that need to span several kilometers in distance.
- The light sources for SMF are usually lasers. Transceivers are often more expensive.
- SMF often operates using wavelengths of 1,310 nm or 1,550 nm, and the cable coating is often colored yellow.

Multimode fiber

- MMF cables are constructed with much larger diameters. For example, one common type of MMF is 62.5/125, meaning that the cable has a diameter of 62.5 µm, compared to the 9 µm of some SMF cables.
- This wider core allows multiple modes of light to propagate through the fiber.
- It limits the maximum link length to much lower distances than SMF.
- Less precise transceivers can be used, allowing the cost of MMF systems to be generally lower than equivalent SMF systems.

Fiber connectors



Fig. – Fiber connectors

Lucent Connector (LC) consists of a small plastic latch, which helps secure the fiber connector to the port, and a 1.25 mm ferrule, which is used to align the fiber optic cable with the connector. It is a small form factor connector, making it suitable for high-density fiber deployments such as in data centers.

Subscriber Connectors (SCs), also known as Square Connectors or Standard Connectors, are connectors that are push-pull square-shaped connectors with 2.5 mm, spring-loaded, ceramic ferrule, and snap-in connector latches. They are easy to disconnect/reconnect and can be found in many network installations.

The Straight Tip (ST) types of connectors consist of a larger ferrule (2.5 mm) and a twist-type, spring-loaded, cylindrical, nickel-plated, or stainless-steel bayonet connector for locking.

Mechanical Transfer Registered Jack (MT-RJ), also known as Media Termination Recommended Jack, are small, duplex fiber connectors, with both fibers terminating on a single 2.45 x 4.4 mm ferrule. They are roughly half the size of SCs and are easy to connect/disconnect from their ports using plastic latches.